

What is a dummy?

Example: Wages and high-school education

Example: Wage by gender

Generalizations and extensions

Time series example

How to predict the future number of passengers?

# ECONOMETRICS

## Lecture 9a

### Binary explanatory variables

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# What is a dummy?

## Slides

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**Dummy** variables—also called **binary** or **indicator** variables—represent qualitative attributes:

- Take value 1 if condition holds, 0 otherwise.

- Examples:

- $$x_i = \begin{cases} 1 & \text{if person } i \text{ is female} \\ 0 & \text{if person } i \text{ is male} \end{cases}$$

- $$x_i = \begin{cases} 1 & \text{if country } i \text{ is part of Europe} \\ 0 & \text{if country } i \text{ is not in Europe} \end{cases}$$

- $$x_i = \begin{cases} 1 & \text{if observation year } i \text{ is 2005} \\ 0 & \text{if observation year } i \text{ is not 2005} \end{cases}$$

# Example: Wages and high-school education

Consider US data on wages and education:

```
# Define link of stata dataset
url <- "http://www.stata.com/data/s4poe5/data/cps5_small.
      dta"
library(haven)           # for reading Stata files
cps <- read_dta(url)     # opens the datafile
head(cps)                # shows some of the data
```

You can see several dummies (black, female, metro, midwest, south, west):

|   | black | educ | exper | faminc | female | metro | midwest | south | wage | west |
|---|-------|------|-------|--------|--------|-------|---------|-------|------|------|
| 1 | 0     | 13   | 45    | 0      | 1      | 1     | 0       | 0     | 44.4 | 1    |
| 2 | 0     | 14   | 25    | 45351  | 1      | 1     | 1       | 0     | 16   | 0    |
| 3 | 0     | 18   | 27    | 91946  | 1      | 1     | 0       | 0     | 15.4 | 0    |
| 4 | 0     | 13   | 42    | 48370  | 0      | 1     | 1       | 0     | 13.5 | 0    |
| 5 | 0     | 13   | 41    | 10000  | 1      | 1     | 0       | 0     | 25   | 1    |
| 6 | 0     | 16   | 26    | 151308 | 1      | 1     | 0       | 0     | 24.0 | 0    |

# Example: Wages and high-school education

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Let us now create our own dummy for high-school education (more than 9 years of schooling):

```
# create a dummy for high school (i.e., educ > 9 years)
cps$highSchool <- ifelse(cps$educ > 9, 1, 0)
```

Our first 6 observations are all individuals with high-school education:

|   | black | educ | exper | faminc | female | metro | midwest | south | wage | west | highSchool |
|---|-------|------|-------|--------|--------|-------|---------|-------|------|------|------------|
| 1 | 0     | 13   | 45    | 0      | 1      | 1     | 0       | 0     | 44.4 | 1    | 1          |
| 2 | 0     | 14   | 25    | 45351  | 1      | 1     | 1       | 0     | 16   | 0    | 1          |
| 3 | 0     | 18   | 27    | 91946  | 1      | 1     | 0       | 0     | 15.4 | 0    | 1          |
| 4 | 0     | 13   | 42    | 48370  | 0      | 1     | 1       | 0     | 13.5 | 0    | 1          |
| 5 | 0     | 13   | 41    | 10000  | 1      | 1     | 0       | 0     | 25   | 1    | 1          |
| 6 | 0     | 16   | 26    | 151308 | 1      | 1     | 0       | 0     | 24.0 | 0    | 1          |

# Example: Wages and high-school education

**How do average salaries differ between those with and without high-school education?**

The answer is provided by following regression

```
model_HS1 <- lm(lwage ~ highSchool, data = cps)
```

The regression yields the result

$$lwage = 2.39 + 0.64 \cdot highSchool \quad (1)$$

which means

$$E[\log(wage)] = \begin{cases} 2.39 & \text{if } highSchool = 0 \\ 2.39 + 0.64 = 3.03 & \text{if } highSchool = 1 \end{cases}$$

Wages are on average 64% higher among those who visited high school.

# Example: Wages and high-school education

## Slides

What is a dummy?

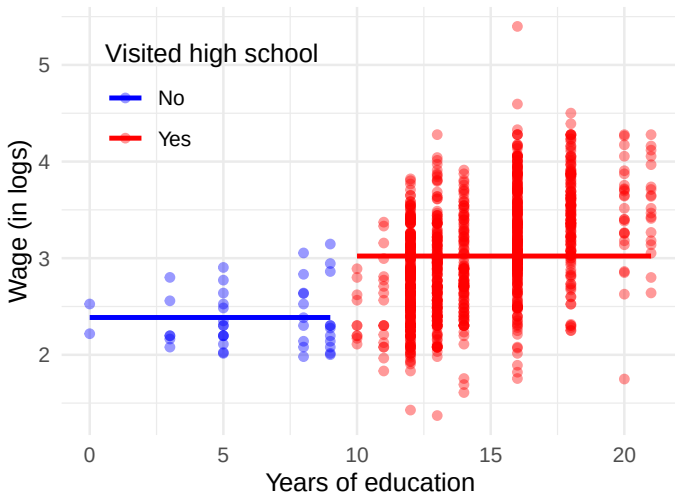
Example: Wages and high-school education

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## How do wages increase per year of education?

The answer is provided by following regression

```
model_HS2 <- lm(lwage ~ educ + highSchool , data = cps)
```

The regression yields the result

$$lwage = 1.72 + 0.11 \cdot educ - 0.26 \cdot highSchool \quad (2)$$

which means that wages increase by 11% per additional year of education.

# Example: Wages and high-school education

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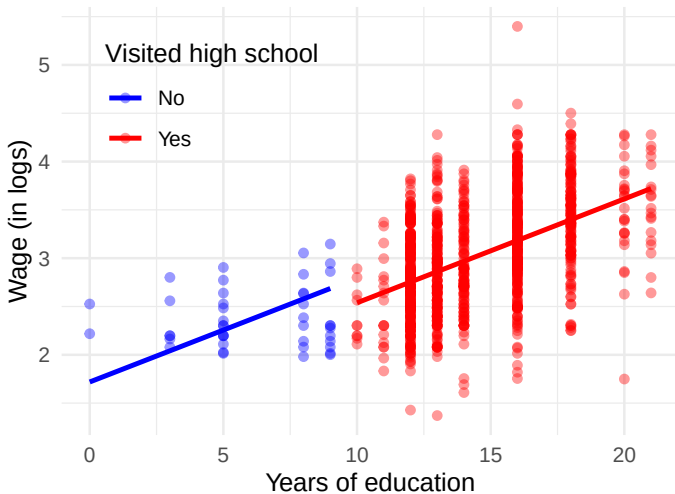
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# Example: Wages and high-school education

**Does the wage increase depend on whether high school was visited?**

The answer is provided by following regression:

```
model_HS3 <- lm(lwage ~ educ + highSchool + highSchool:
  educ, data = cps)
```

which includes the **interaction**  $educ \times highSchool$  as a regressor.

The result is

$$lwage = 2.32 + 0.01 \cdot educ - 0.91 \cdot highSchool \\ + 0.1 \cdot educ \cdot highSchool$$

Thus wages increase by just 1% per year of education for those with no high-school education but by  $1\% + 0.1\% = 11\%$  for those with high-school education.

# Example: Wages and high-school education

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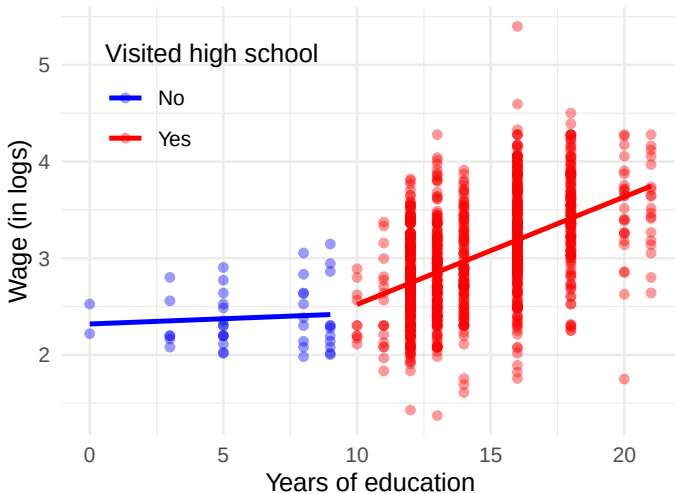
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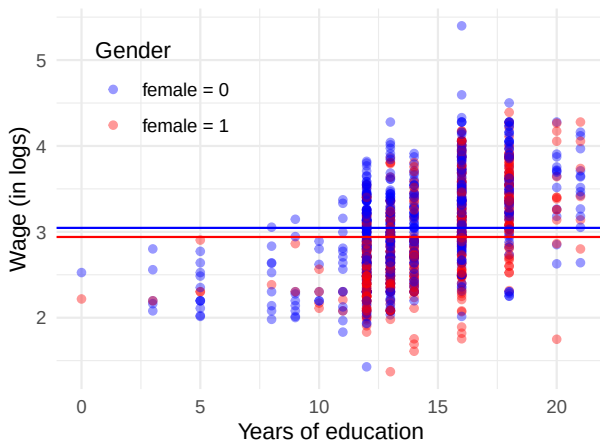
# Example: Wages and high-school education

## Regression Results

|                  | Dependent variable : Log(Wage) |                      |                      |
|------------------|--------------------------------|----------------------|----------------------|
|                  | (1)                            | (2)                  | (3)                  |
| highSchool       | 0.636***<br>(0.085)            | -0.255***<br>(0.089) | -0.912***<br>(0.212) |
| educ: highSchool |                                |                      | 0.101***<br>(0.029)  |
| educ             |                                | 0.108***<br>(0.006)  | 0.011<br>(0.029)     |
| Constant         | 2.386***<br>(0.084)            | 1.719***<br>(0.082)  | 2.319***<br>(0.194)  |
| Observations     | 1,200                          | 1,200                | 1,200                |
| R2               | 0.044                          | 0.263                | 0.270                |
| Note:            | *p<0.1; **p<0.05; ***p<0.01    |                      |                      |

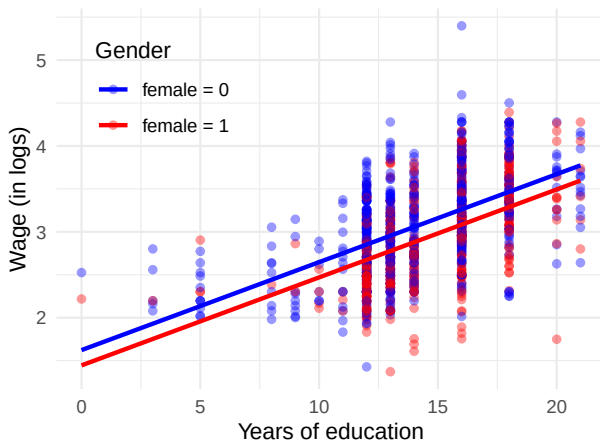
# Example: Wage by gender

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \cdot \text{female}_i + u_i$$



# Example: Wage by gender

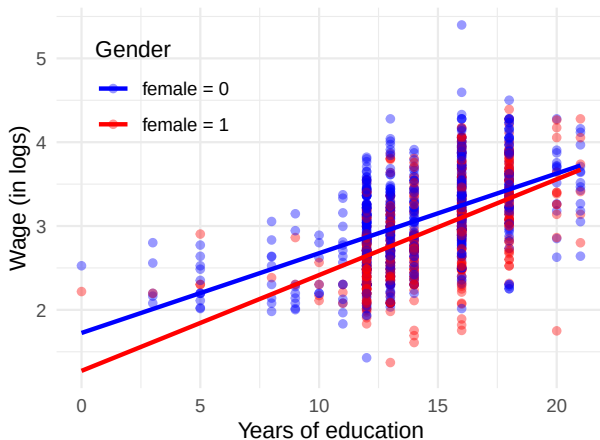
$$\log(\text{wage}_i) = \beta_0 + \beta_1 \cdot \text{female}_i + \beta_2 \cdot \text{educ}_i + u_i$$



# Example: Wage by gender

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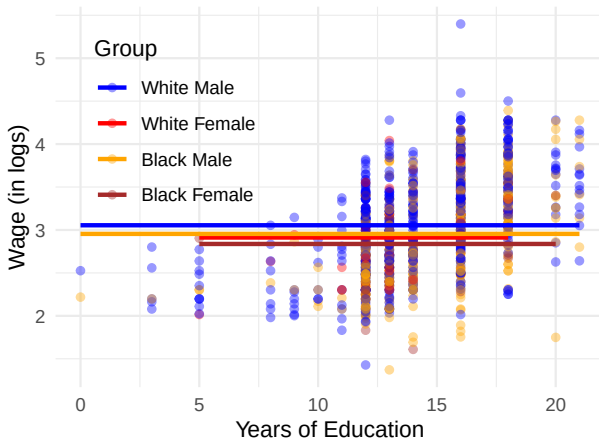
$$\log(\text{wage}_i) = \beta_0 + \beta_1 \cdot \text{female}_i + \beta_2 \cdot \text{educ}_i + \beta_3 \cdot (\text{female}_i \cdot \text{educ}_i) + u_i$$



# Example: Wage by gender

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$$\log(\text{wage}_i) = \beta_0 + \beta_1 \cdot \text{female}_i + \beta_2 \cdot \text{black}_i + \beta_3 \cdot (\text{female}_i \cdot \text{black}_i) + u_i$$



# Example: Wage by gender

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|              | Dependent variable: Log(Wage) |                      |                      |                      |
|--------------|-------------------------------|----------------------|----------------------|----------------------|
|              | (1)                           | (2)                  | (3)                  | (4)                  |
| female       | -0.106***<br>(0.033)          | -0.178***<br>(0.028) | -0.452***<br>(0.144) | -0.101***<br>(0.034) |
| educ         |                               | 0.102***<br>(0.005)  | 0.095***<br>(0.006)  |                      |
| female*educ  |                               |                      | 0.019*<br>(0.010)    |                      |
| black        |                               |                      |                      | -0.142<br>(0.090)    |
| black*female |                               |                      |                      | 0.025<br>(0.117)     |
| Constant     | 3.046***<br>(0.022)           | 1.623***<br>(0.069)  | 1.724***<br>(0.087)  | 3.055***<br>(0.022)  |
| Observations | 1,200                         | 1,200                | 1,200                | 1,200                |
| R2           | 0.009                         | 0.282                | 0.284                | 0.013                |

Note:

\*p<0.1; \*\*p<0.05; \*\*\*p<0.01



# Generalizations and extensions

- You can “dummify” essentially *everything* (cohort, country, religion, tastes, grades, etc.)
- The term country **fixed effects** refers to a dummy for each country in the data (one for Italy, one for Germany, one for France, etc.). You can have FE by all sorts of groups (years, religion, industry, etc.)
- You can also have dummies **as dependent variables**. In that case, the outcome is interpreted as probability.

**Example:** The estimate

$highSchool = 0.95 - 0.01 \cdot black + 0.04 \cdot female$  tells that the **probability** of having high-school education decreases by 1% if the person is black and increases by 4% if the person is female.

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# TIME SERIES EXAMPLE

# How to predict the future number of passengers?

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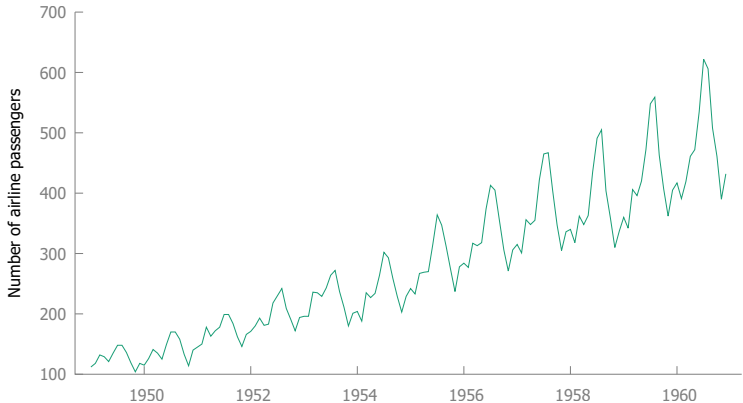
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How to predict the future number of passengers?



# How to predict the future number of passengers?

## Summary

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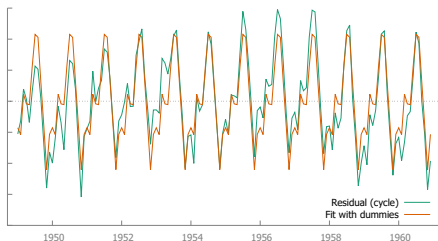
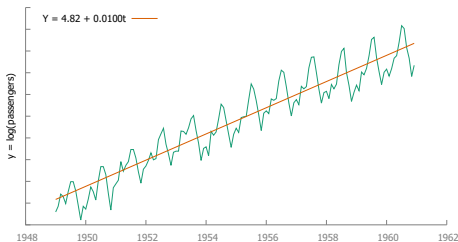
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How to predict the future number of passengers?



- Linear trend + cyclical component.
- Linear trend = slope on  $t = 1, 2, 3, \dots$
- Cyclical component = a dummy for each month

$$y = \beta_1 + \beta_2 t + \delta_1 D_t(\text{January}) + \delta_2 D_t(\text{February}) + \dots + \delta_{11} D_t(\text{November}) + u_t$$

# How to predict the future number of passengers?

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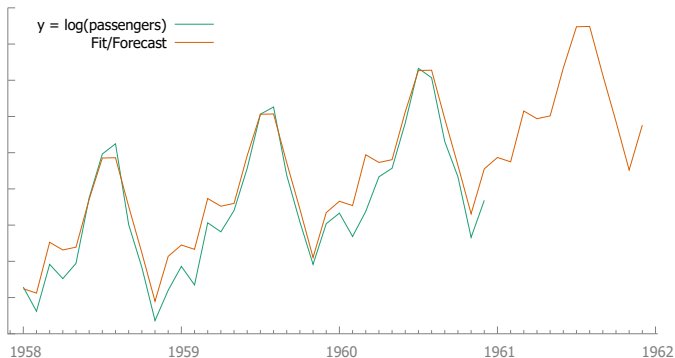
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How to predict the future number of passengers?



$$E(y|t \text{ is February}) = \hat{\beta}_1 + \hat{\beta}_2 t + \hat{\delta}_2 D_t(\text{February})$$